



Programmer

Task A: Algorithms

Algorithm 1

Step 1: Take two positive numbers, A and B .

Step 2: Draw a table with two columns. Put A at the top of the left column and B at the top of the right column.

Step 3: In the next row, in the left-hand column write the number that is half of A (ignoring remainders) and in the right-hand column write the number that is double B .

Step 4: Repeat Step 3 until you have a 1 in the left-hand column.

Step 5: Delete any rows where the left-hand column has an even number.

Step 6: Find the sum of the numbers left in the right-hand column.

1) Carry out algorithm 1 with the following numbers:

a) $A = 8, B = 10$

8	10
4	20
2	40
1	80

The sum of the remaining numbers in the right-hand column is 80

b) $A = 29, B = 34$

29	34
14	68
7	136
3	272
1	544

The sum of the remaining numbers in the right-hand column is 986

c) $A = 13, B = 17$

13	17
6	34
3	68
1	136

The sum of the remaining numbers in the right-hand column is 221

d) $A = 66, B = 56$

66	56
33	112
16	224
8	448
4	896
2	1792
1	3584

The sum of the remaining numbers in the right-hand column is 3696

2) What does algorithm 1 do?

The algorithm calculates the product of the two numbers.



3) Write an algorithm for making a drink of squash or a cup of hot chocolate (more challenging). Give your algorithm to a partner and let them try to 'break' your algorithm. Is there a way for them to misinterpret your instructions?

E.g. There were lots of possibilities here. Hopefully you had lots of fun 'breaking' algorithms. Here are some examples of ways you could have done this:

'Fill the kettle with water' - this could mean put in water until it reaches the very top of the kettle (water will escape when the kettle boils)

- where does the water come from? (tap, toilet?!)

Task B: Logic

1) 'If it is raining, I will use an umbrella.'

What can you conclude if:

- | | |
|--------------------------------|--------------------------------|
| a) It is raining. | <i>I will use an umbrella.</i> |
| b) It is not raining. | <i>Nothing new.</i> |
| c) I am using an umbrella. | <i>Nothing new.</i> |
| d) I am not using an umbrella. | <i>It is not raining.</i> |

2) At Oak National Academy, it is logic day. On logic day, all pupils either always tell the truth or always lie.

You meet Jacob and Jun. Jacob says, 'Exactly one of us is telling the truth today.'

Is it possible to state whether Jacob is telling the truth or not?

No. We can work out that Jun must be lying today but we don't know whether Jacob is telling the truth or not.



3) At Oak National Academy, it is logic day. On logic day, all pupils either always tell the truth or always lie.

You meet Laura and Lucas. Laura says, 'I am lying today **and** Lucas is telling the truth today.'

What can you deduce, if anything, about Laura and Lucas?

Laura and Lucas are both lying today.

For Laura to be telling the truth, every part of her statement must be true. One part being a lie is enough to know she is lying today.

4) At Oak National Academy, it is logic day. On logic day, all pupils either always tell the truth or always lie.

You meet Sam and Sofia.

Sam says, 'One of us is lying and one is telling the truth.'

Sofia says, 'We are both doing the same thing **and** I am lying.'

What can you deduce, if anything, about Sam and Sofia?

Sam is telling the truth and Sofia is lying.